

FUAD AHMED (TUBA, FUAD AHMED)

Candidate for Erasmus Mundus Joint Master in Green Networking & Cloud Computing (GENIAL)

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🎯 ACADEMIC PROFILE & RESEARCH OBJECTIVE

Software systems engineer with 4+ years of professional backend, distributed computing, and Linux systems experience, holding a BSc in Computer Science from St. Mary's University (GPA: 3.20/4.00, English MOI verified). Dedicated to advancing sustainable, low-carbon ICT infrastructures by researching distributed resource scheduling, energy-efficient edge computing, and adaptive telemetry within hyperscale and heterogeneous cloud networks.

🎓 HIGHER EDUCATION

Bachelor of Science (BSc) in Computer Science

Oct 2018 – Aug 2022

Faculty of Informatics, St. Mary's University — Addis Ababa, Ethiopia (Accredited by HERQA)

- **Academic Standing:** Cumulative GPA: **3.20 / 4.00** (~80% academic equivalent, Upper Division).
- **Language of Instruction:** 100% English. Official Medium of Instruction (MOI) Certificate verified.
- **Key Coursework:** Distributed Systems, Operating Systems Architecture, Computer Networks & Protocols, Computer Architecture, Advanced Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Database Management Systems, System Analysis and Design, Probability & Statistics, Discrete Mathematics.
- **Undergraduate Capstone Thesis:** *Design and Implementation of a High-Concurrency Fault-Tolerant Distributed Data Synchronization Protocol* — Architected a lightweight state-synchronization protocol across distributed nodes with optimized payload serialization and low-overhead network heartbeats.

💼 PROFESSIONAL EXPERIENCE

Lead Backend & Distributed Systems Engineer

2022 – Present

Enterprise Technology Solutions & Independent Consultancy — Addis Ababa, Ethiopia

- Architected and deployed high-throughput backend services and microservices handling concurrent workloads using Python (FastAPI/AsyncIO), Go, and Node.js.
- Engineered distributed caching topologies utilizing Redis clusters, reducing primary database load by 48% and mitigating tail read latency in bandwidth-constrained network topologies.
- Designed event-driven message architectures utilizing Kafka and RabbitMQ to decouple asynchronous processing pipelines, ensuring 99.9% uptime during traffic surges.
- Containerized heterogeneous microservices with Docker and orchestrated deployment topologies on Linux environments, optimizing compute resource density and power utilization.

Software Systems Engineer

2020 – 2022

Full-Stack & Systems Development — Addis Ababa, Ethiopia

- Developed transactional relational database schemas and indexed query execution plans on PostgreSQL and MySQL for enterprise data management.
- Implemented secure authentication, authorization, and cryptographic token lifecycles (OAuth2/JWT) across distributed client-server APIs.
- Configured Linux server environments, systemd daemon automations, firewall policies, and automated CI/CD continuous delivery pipelines with Git.

🔬 RESEARCH & TECHNICAL PROJECTS

- **Autonomous Edge-Cloud Telemetry & Automation Engine:** Built a modular system automation daemon in Python/Bash featuring automated kernel resource management, wake-lock telemetry, and headless remote-debugging interfaces under restricted compute footprints.
- **High-Throughput Concurrency Benchmarking Suite:** Developed benchmarking tools in Go to evaluate connection pooling, context switching overheads, and network socket recycling under simulated packet degradation.

- **Open-Source Scholarship Dossier & CI/CD Tracker:** Built an automated tracker with zero-dependency Python CLI tooling and static web dashboard synchronized via GitHub Actions.

TECHNICAL COMPETENCIES

Distributed & Cloud:	Distributed Architectures, Microservices, Edge Computing, Containerization (Docker), RESTful APIs, gRPC, Message Queues (Kafka, RabbitMQ), Redis Caching.
Programming Languages:	Python, Go (Golang), Java, C++, JavaScript/TypeScript, SQL, Bash scripting.
Operating Systems & Net:	Linux (Ubuntu, Debian, kernel tuning), TCP/IP protocols, Network Sockets, X11/Xvfb, Nginx reverse proxying, systemd.
Databases:	PostgreSQL, MySQL, Redis, MongoDB.

LANGUAGES & COMMUNICATION

- **English:** Fluent / Full Professional & Academic Proficiency (Official St. Mary's English MOI Certificate).
- **French:** Elementary (A1) — Actively pursuing Alliance Éthio-Française coursework.
- **Amharic & Afaan Oromo:** Native / Bilingual proficiency.

REFEREES

- **Dr. Tesfaye Assefa** — Assistant Professor, Faculty of Informatics, St. Mary's University.
- **Mr. Birhanu Getachew** — Senior Lecturer & Capstone Advisor, Department of Computer Science, St. Mary's University.